HUARAVITO LTEMS

Long-Term Ecological Monitoring Site

Date: February 28, 2026

Location: (10.104583, -84.564639)

Piedades Sur, San Ramón, Costa Rica

Reserva Huaravito Long-Term Ecological Monitoring Site (LTEMS)

Technical Note

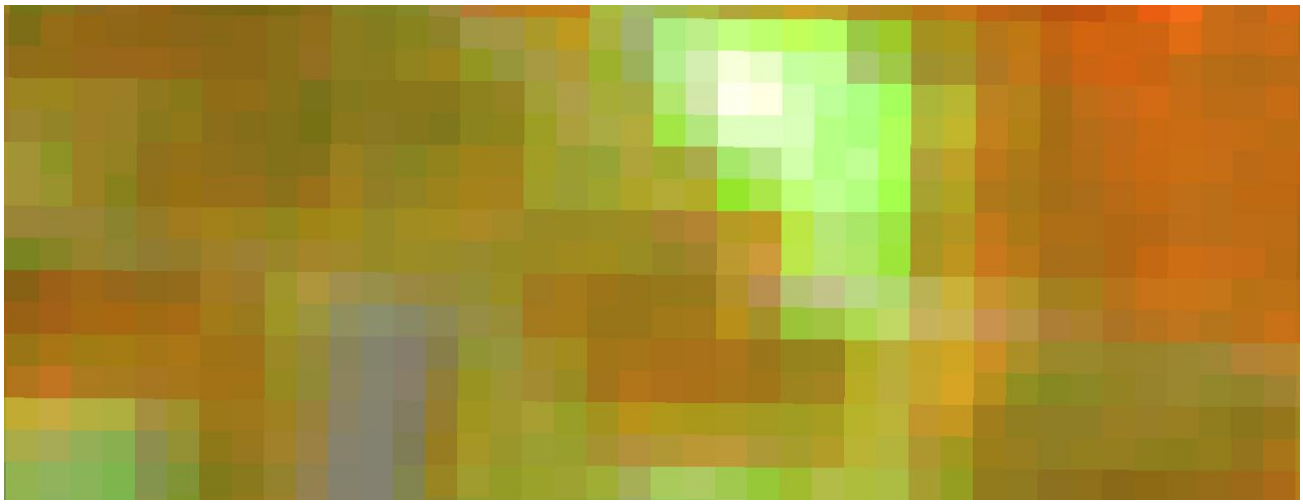
OpenForis Spatial Assessment – Landscape Structural Contrast

1. Objective

To document the structural contrast between a degraded anthropogenic site (tajo) and the adjacent mature forest core (Toyopán) using OpenForis spatial imagery, and to correlate these observations with field-measured thermal gradients recorded during a midday transect on 27 February 2026.

2. Spatial Observation (OpenForis Visualization)

OpenForis imagery reveals a clear structural discontinuity within the Huaravito landscape, with two contrasting sectors identifiable by their spectral and textural signatures:



Cleared Sector (Tajo)

- High surface reflectance consistent with exposed mineral substrate
- Minimal or absent canopy cover
- Fragmented and discontinuous vegetation structure

Forest Core (Toyopán Patch)

- Dense, continuous canopy cover
- Dark-green spectral signature indicative of high chlorophyll density
- Structural layering and high vertical complexity

This contrast indicates a strong difference in canopy density and surface cover between both sectors, and provides a spatial framework for interpreting field-measured thermal data.

3. Field Correlation — 27 February 2026 Transect

Temperature measurements were recorded during peak solar radiation conditions along a transect spanning the degraded zone and forest core:

Zone	Site	Structural Description	Measured Temp.
A	**Tajo (degraded quarry)**	High reflectance, no canopy, exposed substrate	**34.5°C**
B	**Forest edge / transition**	Intermediate structural density, likely edge vegetation	**27–29°C**
C	**Toyopán forest core**	Closed canopy, high vertical complexity, low reflectance	**24.5°C**

Maximum recorded ΔT: 10.0°C (Tajo vs. Toyopán core, peak radiation conditions)

The spatial discontinuity observed in OpenForis corresponds directly with the thermal gradient measured during the midday transect. Measurement time window should be recorded precisely in future transects to strengthen temporal comparability.

4. Structural Interpretation

The imagery and field data jointly suggest the following:

- Areas lacking canopy exhibit higher surface exposure and elevated radiative load.
- The Toyopán forest patch maintains canopy continuity sufficient to significantly attenuate incoming solar radiation.
- Structural complexity likely contributes to reduced surface heating and increased evapotranspiration — though it should be noted that evapotranspiration affects surface temperature more directly than air temperature, and future sensor deployments should distinguish between these where possible.

Critically, the spatial pattern in the imagery suggests that forest-mediated cooling is not a point phenomenon. The structural contrast visible across the landscape indicates that thermal buffering operates at the scale of the entire Toyopán patch, supporting a landscape-scale interpretation of the gradient.

However, while spatial structure aligns with observed thermal differences, long-term synchronized measurements are required before causal relationships can be formally established.

5. Ecological Implications

The forest core may function as:

- A localized thermal buffering zone capable of maintaining temperatures significantly below the surrounding anthropogenic matrix.
- A potential microclimatic refuge — a working hypothesis consistent with the literature on remnant forest fragments in degraded tropical landscapes, pending validation through multi-point sensor data.

The degraded sector (tajo) represents a zone of elevated thermal exposure and reduced structural complexity, consistent with the well-documented thermal penalties of deforestation in premontane environments.

6. Methodological Note

This brief integrates three complementary data sources: remote spatial visualization via OpenForis, a single-day field temperature transect conducted under peak solar conditions, and visual structural assessment. While this convergence of evidence is informative, the following are required for rigorous validation:

- Continuous sensor deployment (multi-point, synchronized logging)
- Precise transect time-window documentation
- Canopy cover quantification (%) and distinction between surface and air temperature measurements
- NDVI or equivalent vegetation index extraction from satellite imagery
- Repeated measurements across seasons to assess temporal stability of the gradient

7. Supplementary Note: Spatial-Thermal Pattern Analysis

Visual inspection of the OpenForis imagery identifies three structurally distinct spatial domains within the study landscape, each corresponding to a thermal zone observed during the field transect:

Zone A — High-Reflectance Sector (upper right and lateral margins)

Characterized by elevated surface reflectance and minimal canopy cover, consistent with the degraded quarry site. Field measurements at this zone recorded surface temperatures of 34.5°C under peak radiation conditions.

Zone B — Intermediate Green Sector (central transition)

Displays intermediate structural density, consistent with edge vegetation or secondary growth. This zone likely corresponds to the transitional thermal range of 27–29°C observed along the measurement transect.

Zone C — Deep Structural Core (high-density canopy)

Dense, continuous canopy with low reflectance and high vertical complexity. Field measurements at this zone recorded 24.5°C, representing the thermal minimum of the transect.

A key observation emerging from this spatial analysis is that the recorded thermal gradient does not represent a point phenomenon. The structural contrast visible in the imagery suggests that forest-mediated cooling operates across a coherent landscape patch rather than at isolated measurement points. This spatial continuity strengthens the working hypothesis of landscape-scale microclimatic buffering as a systemic property of the Toyopán fragment — a hypothesis that will require multi-point, continuous sensor data to formally evaluate.

8. Conclusion

OpenForis imagery confirms the existence of a marked structural contrast within the Huaravito landscape. This structural discontinuity aligns spatially and quantitatively with the 10.0°C thermal gradient recorded during peak radiation conditions on 27 February 2026.

The convergence of remote spatial data and field temperature measurements strengthens the rationale for continued implementation of the Sentinel Microclimate Monitoring Protocol under the Huaravito LTEMS framework, and justifies the deployment of continuous sensor arrays to formally characterize the thermal buffering properties of the Toyopán forest fragment.